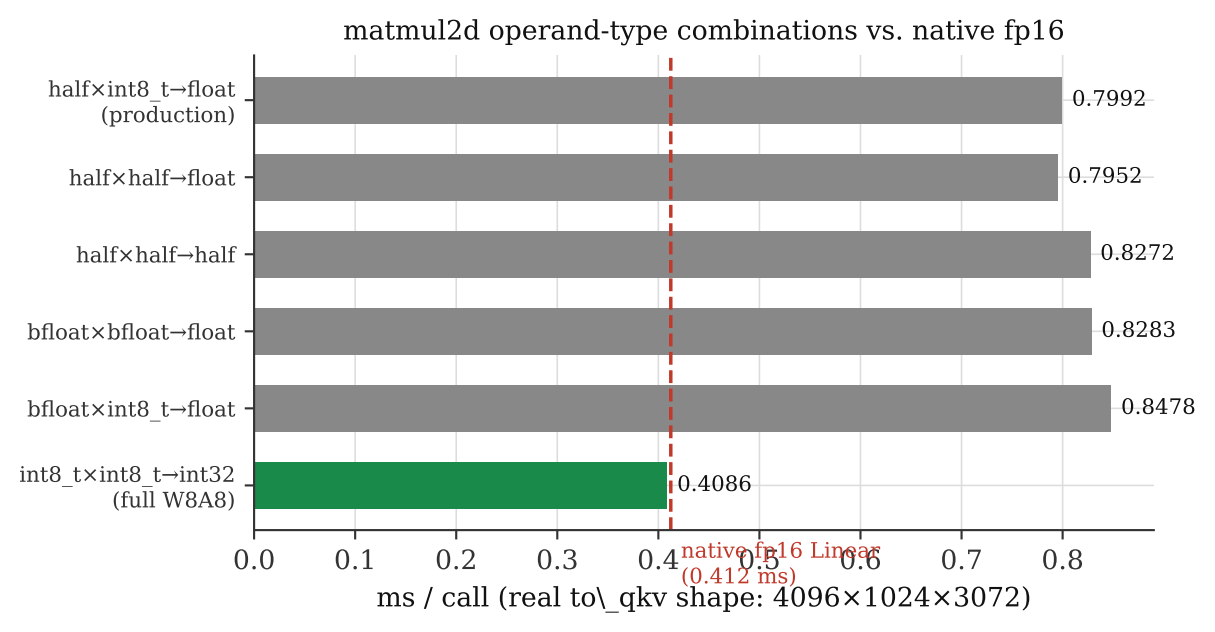
matmul2d operand-type combinations vs. native fp16
half×int8_t→float (production)
half×half→float
half×half→half
bfloat×bfloat→float
bfloat×int8_t→float
int8_t×int8_t→int32 (full W8A8)
0.7992
0.7952
0.8272
0.8283
0.8478
0.4086
native fp16 Linear (0.412 ms)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
ms / call (real to_qkv shape: 4096×1024×3072)